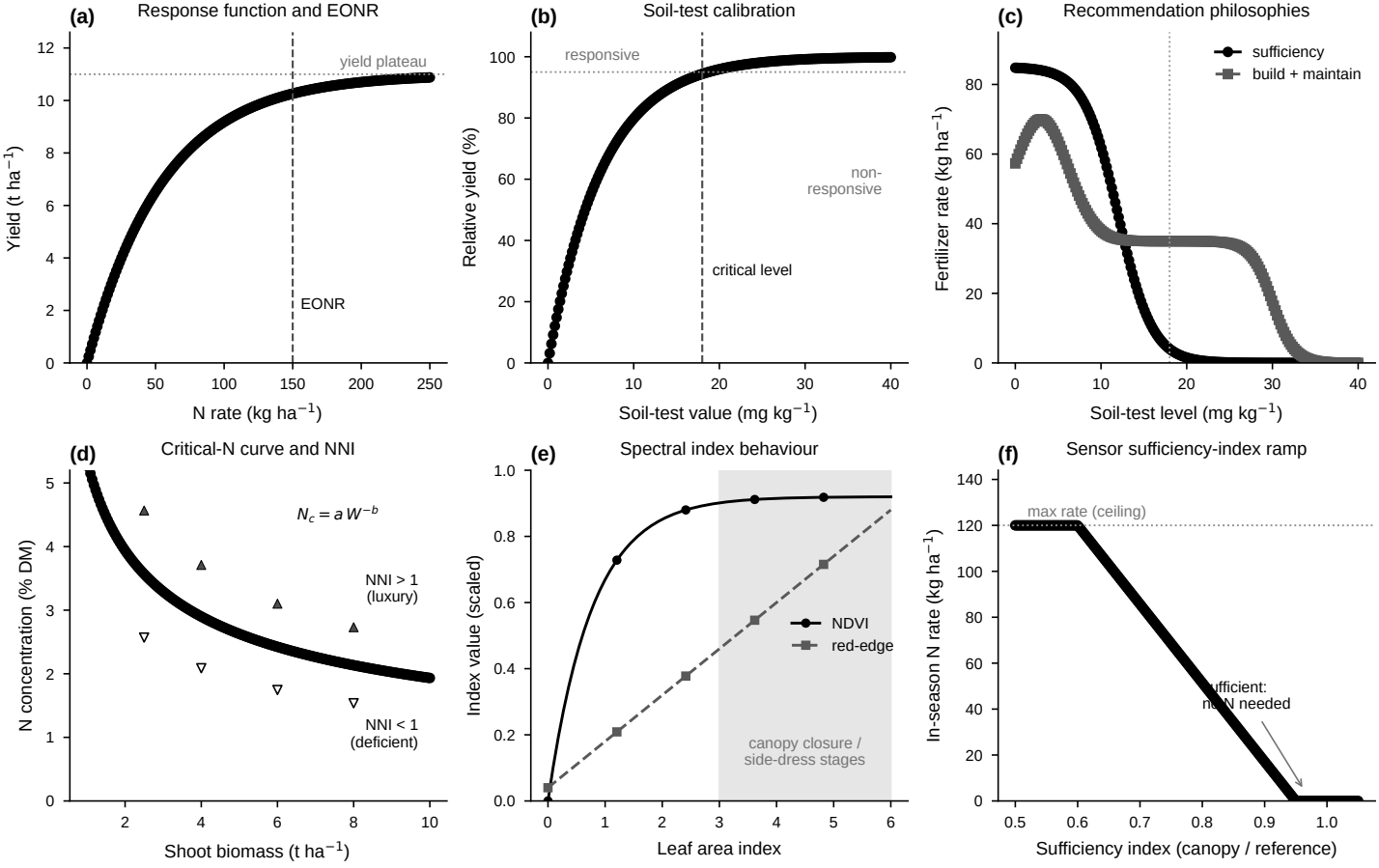
(a)
Response function and EONR
yield plateau
EONR
Yield (t ha^{-1})
N rate (kg ha^{-1})
(b)
Soil-test calibration
responsive
non-
responsive
critical level
Relative yield (%)
Soil-test value (mg kg^{-1})
(c)
Recommendation philosophies
sufficiency
build + maintain
Fertilizer rate (kg ha^{-1})
Soil-test level (mg kg^{-1})
(d)
Critical-N curve and NNI
$N_c = a\,W^{-b}$
NNI > 1
(luxury)
NNI < 1
(deficient)
N concentration (% DM)
Shoot biomass (t ha^{-1})
(e)
Spectral index behaviour
NDVI
red-edge
canopy closure /
side-dress stages
Index value (scaled)
Leaf area index
(f)
Sensor sufficiency-index ramp
max rate (ceiling)
sufficient:
no N needed
In-season N rate (kg ha^{-1})
Sufficiency index (canopy / reference)